Examiner only

6. **Image 6.1** shows a BATAK machine being used to measure an individual's reactions.

Image 6.1

The method for using the BATAK machine is:

- The lights in the buttons flash randomly, for differing lengths of time.
- The individual has to hit each button when it flashes.
- The number of lights successfully hit in 1 minute is recorded.

Table 6.2 shows the results recorded for five year 11 students. Use the data to answer the questions that follow.

Table 6.2

Name	Number of lights hit successfully in 1 minute			
	Attempt 1	Attempt 2	Attempt 3	Mean
Sean	31	34	39	35
Mark	36	38	41	....
Eleri	40	41	42	41
Gwen	29	35	40	35
Gruffydd	21	22	23	22

All the students except Gruffydd did their test during the first lesson of the day. Gruffydd did his test in the last lesson of the day.

(a) Calculate the mean number of lights hit in 1 minute by **Mark**.
Write your answer to the nearest whole number in Table 6.2. [1]
Space for working

 (3400U40-1)

Examiner only

(b) State the name of the student with the fastest reactions. [1]

....

(c) Suggest why Gruffydd's results are not valid in this investigation. [1]

....

....

(d) State **two** other variables that should have been controlled in this method. [2]

....

....

(e) State **two** properties of reflex actions. [2]

....

....

7

Turn over.